WEEK 3 (Mon. / Thu.)

BEFORE YOU START — WHAT TO COLLECT

Inorganic Experiment 1, Session II — control study without catalyst

Tick as you collect. Everything that touches the reaction must be **washed and dried** first — see the box at the foot of the page.

A. Glassware and apparatus

☐ 100 mL round-bottom flask — reaction
☐ 150 mL round-bottom flask — filtrate
☐ Magnetic stirrer bar (small) + retriever
☐ Water-cooled reflux condenser
☐ Rubber tubing × 2, for condenser water
☐ $CaCl_2$ guard tube
☐ Fresh $CaCl_2$ + cotton wool plug
☐ Retort stand + **2** clamps and bosses
☐ Hotplate stirrer
☐ Silicone oil bath + bath thermometer
☐ Hirsch funnel + filter paper of matching size
☐ Filter (suction) flask
☐ Vacuum trap + thick-walled tubing
☐ 2 more clamps for the filtration set-up
☐ Gas adapter, for drying the oil under vacuum
☐ Measuring cylinders, 10 mL and 100 mL
☐ Micropipette **30–200 µL** + tips
☐ Glass dropper / Pasteur pipettes, clean and dry
☐ Spatula, weighing boat or paper
☐ NMR tube + cap + label sticker

B. Reagents and consumables

☐ Dried acetone, **8 mL** (take a little extra)
☐ 1-Phenylethanol, **0.06 mL**
☐ K_2CO_3, **0.07 g**
☐ *n*-Hexane, **∼65 mL** (50 quench + 2 × 5 rinse)
☐ Celite
☐ Mesitylene, **0.07 mL** — internal standard
☐ $CDCl_3$, **0.6 mL**
☐ Silicone oil for the bath
☐ Acetone for washing glassware afterwards
☐ Ice, if the quench runs warm

C. Shared equipment — plan around the queue

☐ Analytical balance (0.07 g needs four decimals)
☐ **Rotary evaporator** — the bottleneck; start the reflux early
☐ Vacuum line or pump for filtration and drying
☐ NMR submission point, and today's cut-off time

D. Personal and paperwork

☐ Own lab coat and safety goggles
☐ Full-length trousers, covered shoes, high-cut socks
☐ Gloves (nitrile provided; ask for butyl if handling acetone at length)
☐ This bench sheet and the datasheet, printed
☐ Calculator and a permanent marker
☐ S/N number from the lab officer
☐ Last week's record of solvent batch and conditions

WHAT MUST BE DRY, AND WHY

The filter flask above all. The manual singles it out: it must be washed and dried *completely* before use. Water left in it dissolves into the hexane filtrate, survives the rotary evaporator poorly, and turns up in the NMR as a peak near δ 1.6 sitting close to the region you are trying to integrate.

Solvents come from the dried-solvent dispenser, not the general bottle. The $CaCl_2$ in the guard tube should be free-running, not caked — caked $CaCl_2$ is spent and no longer excludes moisture.

Collect the filtration set-up before the reflux ends. The quench, filtration and rotary evaporation run back to back, and hunting for a Hirsch funnel with 50 mL of hexane already in the flask is how sessions overrun.

요약. 반응에 닿는 기구는 모두 미리 씻어 말린다. 특히 흡인 플라스크는 매뉴얼이 따로 명시할 만큼 중요한데, 남은 물이 헥세인 여액에 섞여 들어가 NMR에서 δ 1.6 부근에 나타나고 적분하려는 영역과 가깝다. 용매는 일반 병이 아니라 건조 용매 디스펜서에서 받고, 건조관의 $CaCl_2$가 굳어 있으면 이미 수명이 다한 것이므로 새것으로 바꾼다. 환류가 끝나기 전에 여과 장치를 미리 챙겨 둔다. 퀜칭부터 농축까지는 연속으로 이어지므로, 헥세인을 이미 부은 상태에서 Hirsch 깔때기를 찾으러 다니면 시간이 무너진다. 로터리 증발기는 공용이라 대기가 생기니 환류를 일찍 시작하는 편이 낫다.

WEEK 3 (Mon. / Thu.)

INORGANIC EXPERIMENT 1 — SESSION II

Control Study: Oxidation of 1-Phenylethanol *without* Catalyst

Venue	General & Synthetic Chemistry Lab, Block S5 Level-1	**Instrument / technique**	^{1}H NMR, use of micropipette
Time	10:00 a.m. – 4:00 p.m. (Mon. or Thu.)	**Source**	Manual v0724 + lecture notes

PURPOSE AND SAFETY

This run is the baseline for next week's catalysed run, and **the comparison is only valid if the two are matched** — same reflux duration, same solvent batch, same practice. Record anything that differs. Without ruthenium the only oxidant is adventitious atmospheric O_2 admitted during assembly; K_2CO_3 deprotonates but cannot oxidise, so any acetophenone here is background.

Acetone, *n*-hexane and mesitylene are highly flammable; hexane and $CDCl_3$ are toxic and irritating. 1-Phenylethanol, acetophenone, K_2CO_3, Celite and $CaCl_2$ are harmful. Keep solvents away from the hotplate. Fume cupboard, lab coat, goggles, gloves, covered shoes. Used **Celite → designated container**; chlorinated and non-chlorinated solvents separately.

Procedure

1. **8 mL** dried acetone and a stirrer bar into a 100 mL RBF.
2. By micropipette, **0.06 mL** 1-phenylethanol, with stirring.
3. **0.07 g K_2CO_3**, with stirring.
4. Fit the water-cooled condenser with the $CaCl_2$ guard tube *quickly*.
5. **Reflux 60 min.** Oil bath < **100 °C**; acetone boils at 56 °C, so a modest bath is enough. Set it once and leave it.
6. Cool to room temperature. **Cool before the next step.**
7. Add **50 mL *n*-hexane** with stirring to quench.
8. Filter through **Celite** on a Hirsch funnel; collect the filtrate. The suction flask must be washed and *dried completely* beforehand. Pack a thin layer of Celite onto the filter paper *with the suction already running*.
9. Transfer the filtrate to a 150 mL RBF. Rinse the suction flask with *n*-hexane (**2 × 5 mL**) and combine.
10. Remove the solvent on the rotary evaporator.
11. Fit a gas adapter; dry the oily product under vacuum **15–20 min**.
12. By micropipette, **0.07 mL mesitylene**, with stirring. *Internal standard — see Appendix D. It goes in only now, after the solvent is gone.*
13. Add **0.6 mL $CDCl_3$**, transfer the whole solution into a labelled NMR tube with a clean glass dropper, submit for **^{1}H NMR**.

BENCH CAUTIONS

Adding hexane to a hot mixture makes it boil out of the flask. Cool first.

Celite retains residues that would otherwise keep reacting during concentration on the rotary evaporator. Both flasks in the filtration must be clamped — see Appendix A2.

The residue is an impure oil, so no yield can be weighed from it. The conversion comes only from the NMR integration against mesitylene, which means **every transfer from the quench onwards must be quantitative**. There is no point recording the flask mass before the rotary evaporator.

Micropipette: depress to the first stop, immerse 2–3 mm below the surface, release the plunger smoothly, then dispense to the second stop. At 0.06 mL a sloppy technique is a several-percent error in the very quantity the experiment measures.

Before you leave: ☐ NMR tube labelled with name, session and S/N no. ☐ Submitted at the NMR point ☐ Celite discarded correctly ☐ Datasheet signed by GTA

3주차 (월 / 목)

무기 실험 1 — 세션 II

대조 실험: 촉매 없이 1-phenylethanol 산화

장소	General & Synthetic Chemistry Lab, Block S5 Level-1	**장비**	^{1}H NMR, micropipette
시간	오전 10시 – 오후 4시 (월 또는 목)	**출처**	매뉴얼 v0724 + 강의노트

목적과 안전

이 실험은 다음 주 촉매 반응의 기준선이며, **두 세션의 조건이 일치해야만 비교가 성립한다.** 환류 시간, 용매 배치, 조작 방식을 같게 하고 달라진 점이 있으면 반드시 기록한다. 루테늄이 없으면 산화제는 장치를 조립하는 동안 들어온 대기 중 O_2뿐이다. K_2CO_3는 탈양성자화만 할 뿐 스스로 산화시키지 못하므로, 여기서 나오는 acetophenone은 전부 배경 반응의 결과다.
아세톤, n-헥세인, 메시틸렌은 인화성이 높고 헥세인과 $CDCl_3$는 독성·자극성이 있다. 1-phenylethanol, acetophenone, K_2CO_3, Celite, $CaCl_2$는 유해하다. 용매를 핫플레이트에서 멀리 둔다. 흄후드에서 실험복, 고글, 장갑, 발등을 덮는 신발을 착용한다. 사용한 **Celite는 지정 용기에** 버리고, 할로겐 용매와 비할로겐 용매는 분리한다.

실험 절차

1. 건조 아세톤 **8 mL**와 교반자를 100 mL 둥근바닥플라스크에 넣는다.
2. 마이크로피펫으로 1-phenylethanol **0.06 mL**를 교반하면서 가한다.
3. K_2CO_3 **0.07 g**을 교반하면서 가한다.
4. $CaCl_2$ 건조관을 연결한 수냉식 냉각기를 신속히 장착한다.
5. **60분간 환류한다.** 오일배스는 **100 °C 미만**으로 둔다. 아세톤의 끓는점이 56 °C이므로 그리 높일 필요가 없다. 한 번 맞춘 뒤 바꾸지 않는다.
6. 실온까지 냉각한다. **반드시 식힌 뒤 다음 단계로 넘어간다.**
7. 교반하면서 n-헥세인 **50 mL**를 가해 반응을 정지시킨다.
8. Hirsch 깔때기에 **Celite**를 깔고 감압 여과하여 여액을 받는다. 흡인 플라스크는 미리 씻어 완전히 건조해 둔다. Celite는 감압을 건 상태에서 여과지 위에 얇게 깐다.
9. 여액을 150 mL 둥근바닥플라스크로 옮기고, 흡인 플라스크를 n-헥세인 **5 mL씩 2회** 헹궈 합친다.
10. 로터리 증발기로 용매를 제거한다.
11. 가스 어댑터를 연결하고 유상 생성물을 진공에서 **15–20분간** 건조한다.
12. 마이크로피펫으로 메시틸렌 **0.07 mL**를 교반하면서 가한다. 내부표준이며 부록 D를 볼 것. 용매를 다 날린 지금 넣는 데에 이유가 있다.
13. $CDCl_3$ **0.6 mL**를 가하고 깨끗한 유리 스포이드로 전량을 NMR 튜브에 옮겨 라벨링한 뒤 **^{1}H NMR**에 제출한다.

실험대 주의사항

뜨거운 상태에서 헥세인을 넣으면 끓어 넘친다. 반드시 식힌 뒤에 넣는다.
Celite는 단순 여과보조제가 아니라, 로터리 증발기에서 농축하는 동안 계속 반응할 수 있는 잔사를 붙잡는 역할을 한다. 여과 시 두 플라스크를 모두 클램프로 고정한다. 부록 A2를 볼 것.
잔류물은 불순한 기름이라 무게로 수율을 낼 수 없다. 전환율은 오직 메시틸렌 대비 NMR 적분에서 나오므로, **퀜칭 이후의 모든 이송이 정량적이어야 한다.** 로터리 증발기에 걸기 전 플라스크 무게를 재는 것은 의미가 없다.
마이크로피펫은 첫 번째 스톱까지 누르고 액면 아래 2–3 mm에 담근 뒤 플런저를 천천히 놓았다가 두 번째 스톱까지 밀어 토출한다. 0.06 mL 수준에서는 조작이 거칠면 몇 퍼센트의 오차가 생기는데, 그 값이 바로 이 실험이 측정하려는 양이다.

퇴실 전 확인: ☐ NMR 튜브에 이름·세션·S/N 라벨 ☐ NMR 제출대에 제출 ☐ Celite 지정 용기에 폐기 ☐ 데이터시트에 GTA 서명

DATASHEET — INORGANIC EXPERIMENT 1 — WEEK 3

Session II: Control Study — Oxidation *without* Catalyst

Full name (as on student ID)	Matriculation no.	Date
Lab session ☐ Mon ☐ Thu	Fumehood no.	S/N no. assigned by lab officer

1. Reagents charged

Reagent	Nominal	Actual	Remarks
Dried acetone	8 mL		
1-Phenylethanol	0.06 mL		
K_2CO_3	0.07 g		
n-Hexane (quench / rinse 2×)	50 / 5 mL		
Mesitylene (internal standard)	0.07 mL		
$CDCl_3$	0.6 mL		

2. Reaction conditions

	Nominal	Actual	Remarks
Reflux start / end time	—		
Reflux duration	60 min		
Oil bath temp. / hotplate setting	$< 100\,°C$		
Rotary evaporator bath temp. / pressure	—		
Vacuum drying of oil	15–20 min		

3. Conditions to reproduce in Session III

Write these down now. Next week's comparison depends on matching them.

Solvent batch / bottle	Ambient conditions (w

4. Observations

Colour on charging, during reflux and after cooling; appearance of the Celite pad and of the filtrate; appearance of the oily residue after concentration.

5. NMR sample

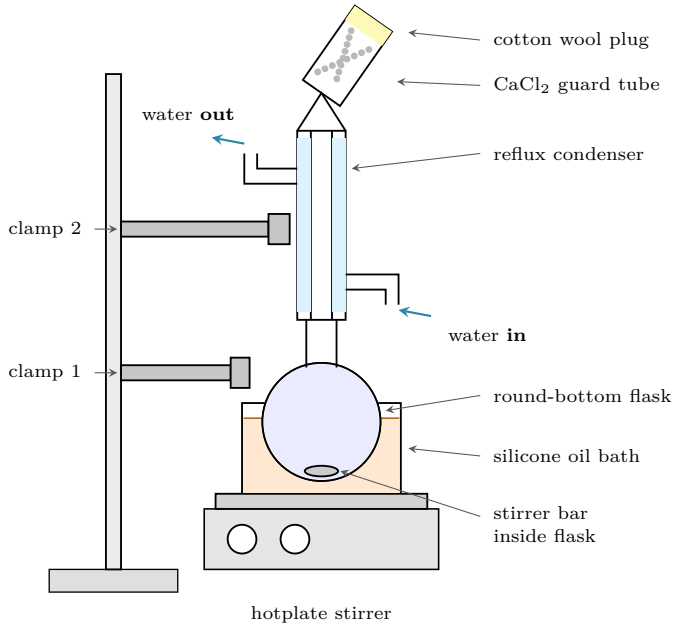
☐ Tube labelled with name, session and S/N no. ☐ Submitted S/N no. _____

To be completed at the end of the session		
Student signature	GTA name	GTA signature

Raw data and observations must be signed by the GTA before leaving the laboratory. This datasheet is submitted as an appendix to the report.

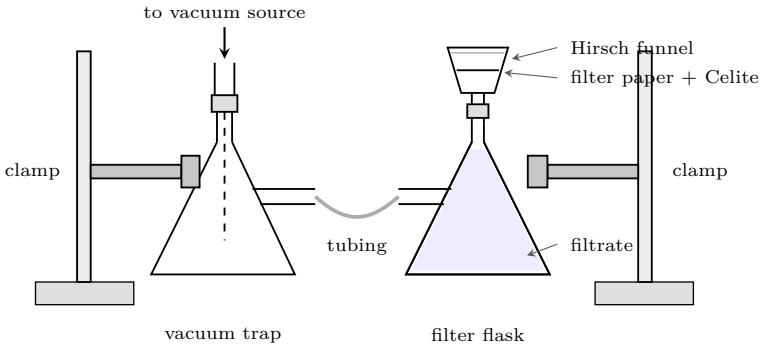
Appendix A — Apparatus Setup

A1. Moisture-free reflux assembly



Three things this drawing is meant to fix. Cooling water enters at the *bottom* and leaves at the *top*, so the jacket stays full; reversing it leaves an air pocket and the condenser stops working. **Clamp 2** holds the condenser and **clamp 1** holds the flask neck, so that neither rests its weight on the oil bath. Both grip the *necks*, and they are snug rather than forced: glass that is being heated expands, and a rigid clamp bearing on it will crack it. The grip must still be secure, because a condenser that slips lets cooling water run into the hot oil. The guard tube is packed with $CaCl_2$ and plugged with cotton wool; it excludes moisture while still venting, so it must never be sealed.

A2. Suction filtration with a vacuum trap



Both flasks must be clamped. A filter flask under vacuum tips easily once tubing is attached, and thick-walled glass under reduced pressure fails violently. The trap sits between the flask and the vacuum line so that liquid drawn over cannot reach the pump; it must be *empty* before use, and its inner tube must extend below the side arm or it will not trap anything. For Session I the funnel carries filter paper only; for Sessions II and III a thin layer of Celite is packed on top of the paper *with the suction already running*, so the pad does not lift.

요약. 냉각수는 아래로 들어가 위로 나가야 재킷이 물로 가득 찬다. 반대로 연결하면 공기층이 생겨 냉각이 되지 않는다. 클램프는 플라스크 목과 냉각기를 각각 잡아 무게가 오일배스에 실리지 않게 하되, 가열되는 유리를 힘으로 조이지 않는다. 유리가 팽창하면서 깨진다. 그렇다고 느슨하면 냉각기가 밀려 물이 기름으로 흘러들므로 알맞게 물리는 것이 요령이다. 건조관은 수분을 막되 압력은 빠져나가야 하므로 절대 밀봉하지 않는다. 감압 여과에서는 여과 플라스크와 트랩을 모두 클램프로 고정하며, 트랩의 안쪽 관은 반드시 곁가지보다 아래까지 내려와 있어야 한다.

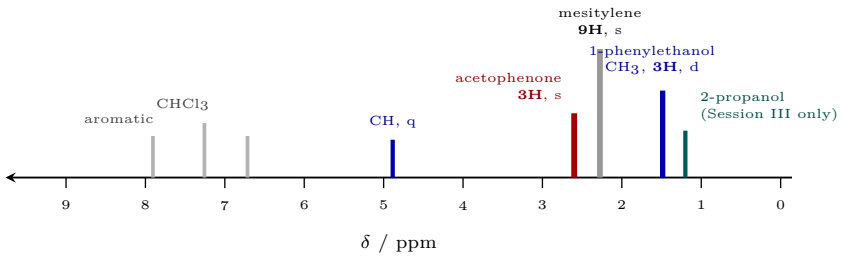
Appendix D — Quantitative ^{1}H NMR with an Internal Standard

D1. Why the yield cannot be weighed

After the rotary evaporator you hold an oily residue of unreacted 1-phenylethanol, any acetophenone formed, traces of solvent and, in Session III, ruthenium residues. Weighing it measures the mixture, not the product. The conversion must therefore come from the ^{1}H NMR spectrum — but an integral is only a *ratio*, and becomes a molar amount only when something of known amount shares the tube. That is what mesitylene is for.

A usable standard must be inert towards everything in the tube, must not overlap the signals being measured, must be added in an accurately known amount, and must survive the sample preparation. Mesitylene qualifies on all four counts: an unreactive arene whose nine equivalent methyl protons give one tall singlet in an otherwise clear region, delivered as 0.07 mL by micropipette, a known 0.503 mmol. This is also why it goes in *after* the solvent is stripped — add it earlier and the rotary evaporator removes an unknown fraction, and the known amount is no longer known.

D2. Where the signals fall



The two signals that matter are well separated: **mesitylene CH_3 near δ 2.3 (9H, singlet)** and **acetophenone $COCH_3$ near δ 2.6 (3H, singlet)**. Unreacted 1-phenylethanol is identified by its methyl *doublet* near δ 1.5 and its methine quartet near δ 4.9. In Session III only, 2-propanol appears as the hydrogen-transfer co-product; its methyl doublet is expected near δ 1.2 in $CDCl_3$, so a doublet at 1.4–1.5 is far more likely to be leftover 1-phenylethanol — check the multiplet positions before assigning it.

D3. The calculation

An integral is proportional to the number of protons, so dividing each integral by the number of protons giving rise to it puts both species on the same molar footing:

$$n_{\text{analyte}} = \frac{I_{\text{analyte}}/H_{\text{analyte}}}{I_{\text{standard}}/H_{\text{standard}}} \times n_{\text{standard}}$$

WORKED EXAMPLE — SUBSTITUTE YOUR OWN INTEGRALS

Set the mesitylene singlet to 9.00 when you integrate. Suppose the acetophenone singlet then integrates to 0.12. With $n_{\text{standard}} = 0.503$ mmol,

$$n_{\text{acetophenone}} = \frac{0.12/3}{9.00/9} \times 0.503 = 0.0201 \text{ mmol},$$

which against the 0.4975 mmol of 1-phenylethanol charged is a conversion of **4.0%**. The same arithmetic with an integral of 0.28 gives 0.0470 mmol and **9.4%**. The difference between the two sessions *is* the result of the experiment.

D4. Things that quietly ruin the number

Material lost in transfer	The standard is referenced against the *whole* residue. Anything left in the suction flask, on the Celite or in the rotary evaporator flask lowers the apparent conversion.
Standard added too early	Mesitylene largely survives the rotary evaporator, but not reliably. Add it afterwards, as the manual specifies.
Incomplete dissolution	Undissolved residue is invisible to NMR. Swirl until the $CDCl_3$ solution is homogeneous.
Overlapping solvent peaks	Residual $CHCl_3$ 7.26, water ~1.56, acetone ~2.17, hexane 0.88 and 1.26. Acetone sits uncomfortably close to the mesitylene singlet.
Careless integration	Integrate over a defined region with flat baseline on both sides, and integrate identically in both sessions or the comparison is meaningless.

Submission requirement. At least one spectrum per session must be presented as a **full spectrum from 0 to 10 ppm with every signal integrated** and the file name visible on it. Integrating only the region of interest will lose marks even if the number is right.

요약. 회수한 유상 잔류물은 불순물이라 무게로 수율을 낼 수 없다. NMR 적분값은 비율일 뿐이므로, 같은 튜브에 양을 정확히 아는 물질이 있어야 몰수로 환산된다. 메시틸렌이 그 역할을 하며, 메틸 9H가 한 봉우리로 깨끗한 영역에 나오기 때문에 적합하다. 용매를 다 날린 뒤에 넣는 이유는 그 전에 넣으면 로터리 증발기에서 얼마가 날아갔는지 알 수 없어 '아는 양'이 아니게 되기 때문이다. 계산은 각 적분값을 해당 수소 수로 나눈 뒤 비를 취하고 표준의 몰수를 곱하면 된다. 퀜칭 이후 모든 이송이 정량적이어야 하며, 두 세션의 적분 방식이 같아야 비교가 성립한다.